

Technology Stack

Date	27/06/2026
Team Id	
Project Name	Smart Lender
Maximum Marks	2 Marks

Technology Stack:

A **Technology Stack** is the combination of programming languages, frameworks, libraries, tools, and software used to design, develop, and deploy the Smart Lender application. It ensures the system functions efficiently, provides accurate loan predictions, and delivers a smooth user experience.

S.No	Architecture Component	Technology Chosen	Justification
1	Frontend	HTML, CSS	Provides a simple, responsive, and user-friendly interface for users to enter loan details and view prediction results.
2	Backend	Python Flask	Handles user requests, processes input data, connects the ML model, and generates prediction results efficiently.
3	Machine Learning Model	Random Forest Classifier	Offers high prediction accuracy, handles structured data effectively, and reduces overfitting compared to a single decision tree.
4	Data Processing	Pandas & NumPy	Used for data cleaning, preprocessing, feature handling, and numerical computations during model development.
5	Model Persistence	Pickle (.pkl)	Saves the trained Machine Learning model and allows it to be loaded quickly without retraining.
6	Development Environment	Visual Studio Code & Jupyter Notebook	VS Code is used for application development, while Jupyter Notebook is used for data analysis, model training, and experimentation.

Additional Technologies:

Category	Technology
Programming Language	Python

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Machine Learning Library	Scikit-learn
Data Analysis	Pandas
Numerical Computing	NumPy
Development Environment	Jupyter Notebook
Version Control	Git & GitHub